

# Resorcinol–formaldehyde resin combining epoxy resin as composite coating with efficient anti-corrosion resistance and anti-biofouling capacity

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## ABSTRACT

In marine environments, microbial contamination and corrosion have persistently threatened the integrity of ships and marine equipment, driving the demand for high-performance coatings integrating anti-corrosion and anti-biofouling functionalities. Herein, for the first time, resorcinol-formaldehyde resin (RF) was used as the filler to introduce epoxy resin (EP) to form an RF/EP composite coating with dual-function anti-corrosion and anti-biofouling protection. Anti-corrosion resistance tests, including open circuit potential (OCP), electrochemical impedance spectroscopy (EIS), and neutral salt spray (NSS), revealed that the RF/EP-6 composite coating exhibits superior anti-corrosion properties, with an impedance radius of  $1.56 \times 10^9 \Omega \cdot \text{cm}^2$ , which surpasses the impedance of pure-phase EP ( $3.64 \times 10^6 \Omega \cdot \text{cm}^2$ ) by more than two orders of magnitude. In addition, the composite coating reflected significant anti-biofouling effects against bacteria (*E. coli* and *S. aureus*) and algae (*Chlorella*), with only a low amount of microorganisms being able to adhere to the surface, which was mainly attributed to the incorporation of RF resin, producing strong oxidative active species ( $\bullet\text{OH}$ ,  $\bullet\text{O}_2^-$  and  $\text{H}_2\text{O}_2$ ). Our study introduces an innovative design concept aimed at developing multifunctional coatings that provide both anti-biofouling and anti-corrosion capabilities for marine equipment field.

## 1. Introduction

Marine resources have enormous economic potential but are currently underdeveloped [1]. To achieve the rapid development of the Marine industry, it is necessary to use Marine equipment, but Marine equipment is currently facing serious corrosion problems [2,3]. Corrosion in the marine environment, especially in facilities such as ships and offshore platforms, is extremely difficult to control. Losses due to equipment failure due to corrosion are unimaginably large and can account for up to one-third of the total corrosion, amounting to about 700 billion RMB [4,5]. Another point is that biological contaminants attached to metal substrates accelerate fuel consumption for ship shipping and further damage to the metal materials [6]. There is thereby an

urgent need but it is still a significant challenge to develop effective anti-corrosion and anti-biofouling technology for Marine equipment.

Carbon steel is an integral part of metallic substances, widely used in ships, Marine equipment and other industries [7]. Currently, the main techniques for preventing metal corrosion include surface treatment and modification of metallic materials, applying anti-corrosion coatings to their surfaces, utilizing electrochemical protection, and loading corrosion inhibitors [8]. Anti-corrosion coatings are the simplest and easiest technology to apply, so a wide variety of coatings are constantly appearing on the market. Epoxy resin (EP) coatings are extensively utilized in marine anti-corrosion studies due to their excellent chemical inertness, strong adhesion, galvanic isolation, and ease of curing [9]. A pure EP coating acts as a unique physical barrier to shield metals from

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corrosion, but a large number of microporous defects and polar channels are formed during the curing process, which makes them unable to provide long-term protection for metals [10]. Besides, pure EP coating cannot display anti-biofouling properties, and the long-term attachment of microorganisms has also seriously affected the anti-corrosion protection performance of the coating, thus the development of efficient coatings with dual functions of anti-corrosion and anti-biofouling properties has become an urgent goal to be achieved in the material industry at present.

In this regard, the incorporation of fillers or functional fillers into EP coatings may be a viable way to make up for its shortcomings. Resorcinol-formaldehyde (RF) resins, as crosslinking polymers, are distinguished by their low cost, simple synthesis process, and high stability [11], enabling them to enhance EP coatings by addressing defects and improving mechanical performance [12]. In 2019, Shiraishi et al. used a high-temperature hydrothermal synthesis method to prepare RF resins, which absorb a broad wavelength of light up to 700 nm and are capable of catalyzing water oxidation and  $O_2$  reduction via photo-generated charges [13]. Under simulated sunlight irradiation, the RF resin produced  $H_2O_2$  with a solar energy conversion efficiency of  $>0.5\%$ , mainly due to its inclusion of  $\pi$ -conjugated and  $\pi$ -stacked benzenoid-quinoid donor-acceptor resorcinol couples. In addition, Du et al. successfully established a photo-self-Fenton system of resorcinol-formaldehyde resin/MIL-88 A(Fe) (RF/88 A) S-scheme heterojunction, which can generate and activate  $H_2O_2$  in situ under visible light irradiation to achieve tetracycline (TC) degradation and sterilization, and can kill about 100 % of *E. coli* and *S. aureus* [14]. Introducing RF resin as a novel functional filler into EP coatings and constructing multifunctional integrated coatings with it, provides an effective design idea for coatings in terms of corrosion protection and anti-biofouling resistance.

In this work, a bifunctional composite coating combining epoxy resin (EP) and RF filler was successfully synthesized and applied for the anti-corrosion and anti-biofouling resistance of Q235 carbon steel. For the anti-corrosion capability of RF/EP composite coating, electrochemical tests are carried out using the electrochemical workstation. Furthermore, the RF/EP composite coating was specifically tested for its resistance to bacteria (*E. coli* and *S. aureus*) and algae (*Chlorella*). In addition, the adhesion of algae on the surface of the RF/EP composite coating was observed using scanning electron microscopy (SEM) for a short period of time. Finally, the corresponding anti-corrosion and anti-biofouling mechanism of the RF/EP composite coating was discussed in detail.

## 2. Experimental

### 2.1. Chemicals

Resorcinol ( $C_6H_6O_2$ ), formaldehyde ( $CH_2O$ ), aqueous ammonia ( $NH_3 \cdot H_2O$ ) and ethanol absolute ( $C_2H_6O$ ) were provided from Shanghai Macklin Biochemical Co., Ltd. (China). Epoxy resin (EP, model number: E-44) and curing agent (model number: T-31) were purchased by Qingdao Yousuo Chemical Co., Ltd. (China).

### 2.2. Specific composition of EP and curing agent

EP (E-44, CAS No. 38891-59-7), color  $\leq 40$  Pt–Co, epoxy equivalent 210–230  $g \cdot mol^{-1}$ , hydrolyzed chlorine  $\leq 0.5\%$ , inorganic chlorine  $\leq 50$   $mg \cdot kg^{-1}$ , volatiles  $\leq 0.6\%$ , the softening point at 14–23 °C. T-31 curing agent is a performance of a comprehensive phenol-formaldehyde amine (PAA) epoxy curing agent, light brown transparent viscous liquid, density (20 °C) of 1.08–1.09  $g \cdot cm^{-3}$ , amine value of 460–550  $mg$  KOH/g, viscosity (25 °C) for 1000–1300 mPa·s, curing agent also contains phenol hydroxyl, primary and secondary amine groups and other active groups.

### 2.3. Preparation of resorcinol-formaldehyde resin

Specifically, 792 mg of  $C_6H_6O_2$  (99 %), 1080  $\mu L$  of  $CH_2O$  and 460  $\mu L$  of  $NH_3 \cdot H_2O$  were added to deionized water (60 mL). After stirring the mixture at room temperature, a white colloidal suspension was formed, transferred into a stainless-steel reaction kettle, and heated at 250 °C for 24 h in a baking oven. Upon cooling to room temperature, the solid powder was obtained through thorough washing with water and absolute ethanol. Finally, obtaining resorcinol-formaldehyde resin (labeled as RF) by vacuum drying at 60 °C for 12 h.

### 2.4. Preparation of RF/EP composite coating on Q235 carbon steel substrate

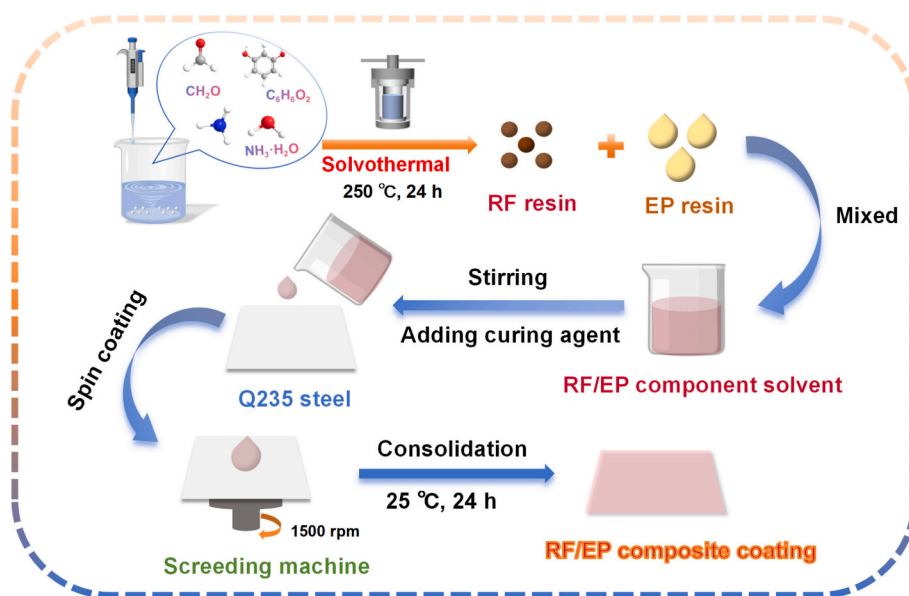
After polishing the surface of the Q235 carbon steel sheet with sandpaper, it was ultrasonically cleaned in anhydrous ethanol solution for 1 h to remove surface stains. Next, the detailed coating preparation process is given as follows (Scheme 1): firstly, RF powder was placed and dispersed in EP jelly; secondly, the curing agent is added to the above dissolvent to ensure that the ratio of the EP to the curing agent was 5:1; thirdly, a small amount of defoaming agent and absolute ethanol were added. After that, the above mixture was stirred for 15 min at room temperature to obtain RF/EP coating liquid. Finally, a certain amount of RF/EP coating solution was poured onto the Q235 steel plate, which was put into a screeding machine for spin-coating (1500 rpm), and solidified at 25 °C for 24 h to obtain the Q235 steel plate wrapped with RF/EP composite coating. Besides, the RF/EP composite coatings were synthesized in three different ratios, as provided in Table 1. For comparison, pure EP coating was also prepared as described above without adding the RF resin. The thickness of all coated samples ranged from 100  $\mu m$  to 110  $\mu m$  (Fig. S1).

### 2.5. Characterizations

The prepared samples were analyzed by Fourier transform infrared spectroscopy (FT-IR; Thermo Scientific Nicolet 6700, USA; the range of test data obtained was 500–4000  $cm^{-1}$ ). X-ray diffraction (XRD, PANalytical B-V, X Pert PRO MPD; using  $Cu K\alpha$  radiation; scanning speed 7°/min) was used to analyze the functional groups and crystal structure characteristics of the prepared samples. The morphology and elemental composition of the prepared samples were examined using a scanning electron microscope (SEM, JSM-7001F, Japan). The surface roughness of the prepared samples was examined by the confocal laser scanning microscope (LEXTOLS4000, Japan). The thermogravimetric analysis (TGA) of the samples was performed using a thermogravimetric analyzer (Pyris Diamond TG-DTA, Japan), where approximately 5 mg of sample was heated at a rate of 15 °C per minute within the temperature range of 0 °C to 600 °C. An electron spin resonance spectrometer (ESR, EPR200M) was used to analyze the prepared samples. The X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha, Al  $K\alpha$  ray) was used to characterize the chemical composition of the prepared materials. The UV–Vis absorption spectra was obtained by UV-2450 spectrophotometer. Transient photocurrent maps of the prepared samples were characterized by the CHI 760E workstation and three-electrode system. A glycercury electrode (4.5 M KCl), prepared samples, and a platinum plate electrode (specification:  $10 \times 10 \times 0.1$  mm) were used as reference, working and counter electrodes, respectively, and 0.5 M  $Na_2SO_4$  aqueous solution was used as the electrolyte. The coating thickness was measured using a Sigma AR932 Composite Coating Thickness Gauge.

### 2.6. Electrochemical impedance spectroscopy test of the as-prepared coatings

To create a corrosive environment, we pasted a plastic cup full of 3.5 wt% NaCl on the Q235 carbon steel with the coating for EIS



**Scheme 1.** A schematic diagram illustrating the preparation process of the RF/EP composite coating on Q235 carbon steel.

**Table 1**

Synthetic raw material of RF/EP composite coatings with different ratios.

| Sample names | EP (g) | RF (mg) | Curing agent (g) |
|--------------|--------|---------|------------------|
| RF/EP-4      | 3      | 4       | 0.6              |
| RF/EP-6      | 3      | 6       | 0.6              |
| RF/EP-8      | 3      | 8       | 0.6              |

measurements (Fig. S2). The detailed procedures of this experiment are as follows: firstly, the top of the plastic cup (top diameter: 4.8 cm, bottom diameter: 3.5 cm, height: 4.1 cm) was glued to the surface of Q235 carbon steel with the coating sample and waited for 24 h for curing. Then, the lower bottom of the plastic cup was cut off, 40 mL of 3.5 wt% NaCl solution was poured into the cup and the samples were placed on a sunny balcony for corrosion testing. After immersion at certain time intervals (1, 14 and 28 days), the corrosion resistance tests were carried out using an electrochemical workstation (CHI 760E). The EIS measurements were performed over a frequency range of 100,000 Hz to 0.01 Hz with an amplitude of 0.01 V.

## 2.7. Adhesive strength test of the as-prepared coatings

The adhesive strength test was carried out using a BGD 500 digital display pull-off method adhesion tester. The spindle (20 mm in diameter) was glued to the surface of the coated samples using DP 810 adhesive, and the test was conducted after waiting for 24 h of curing at room temperature. To ensure precision, each sample underwent three measurements, and the average was calculated to determine the accurate adhesive strength.

## 2.8. Contact angle test for the as-prepared coatings

The water contact angle (WCA) of the coatings was measured using a C311–300 contact angle tester. For the test, 2  $\mu$ L of deionized water was applied to the coating surface at room temperature using the fixation method. Each sample was measured three times, and the average value was recorded to determine the precise contact angle.

## 2.9. Tensile test of the as-prepared coatings

The tensile test was carried out using a WAW-2000Q universal

testing machine. The samples were made into a dumbbell shape (Fig. S3) and tested at room temperature at a tensile rate of 0.5 mm/min.

## 2.10. Neutral salt spray (NSS) test of the as-prepared coatings

To evaluate the corrosion resistance of different coated samples, a neutral salt spray accelerated corrosion test was carried out using a QY-90 A salt spray tester with the test standard GB/T 10125–2021. 5 wt% NaCl was sprayed as the salt solution, and the pH value was 6.7. The coated side of the samples was kept upward during the test. The total duration of the test was 72 h, and changes in the surface of the samples were observed at 24 h intervals.

## 2.11. Anti-biofouling test of the as-prepared coatings

The plate counting method was used to determine the bacterial inhibitory properties of *E. coli* and *S. aureus* on the coated surface (specification: 20  $\times$  20 mm), 1 mL of bacterial solution ( $10^5$  CFU/mL) was first inoculated onto the coated surface, where iron sheets were blanked for control. Then the coated samples were incubated at 37 °C for 3 h. For samples requiring light, 20 min of light was added at the beginning of the incubation, and at the end of the incubation, the coated surfaces were rinsed with PBS solution to ensure that the bacterial solution was diluted 1000 times. Finally, 2  $\mu$ L of each diluted sample was taken as a uniform drop on the surface of the LB agar plate, and the colonies were counted after incubation at 37 °C for 24 h.

The bacterial inhibition rate of the coating was calculated as follows:  $R = (A - B) / A \times 100\%$ , where R is the bacterial inhibition rate, A is the number of colonies in the blank control, and B is the number of colonies after contact with the coating.

Sterilized seawater and chlorella were mixed in a 4:1 ratio in triangular flasks, with nutrient solution added. The flasks were sealed using sterilized kraft paper and twine. Chlorella was incubated under controlled conditions at 25 °C for 72 h with a 14:10 light/dark cycle, and the flasks were shaken three times daily to promote algal growth. At the end of cultivation, the Chlorella suspension was diluted to  $10^5$  CFU/mL and incubated with the coating samples in 24-well plates for 24 h. Algal coverage was evaluated using SEM to observe Chlorella attachment on the surface.

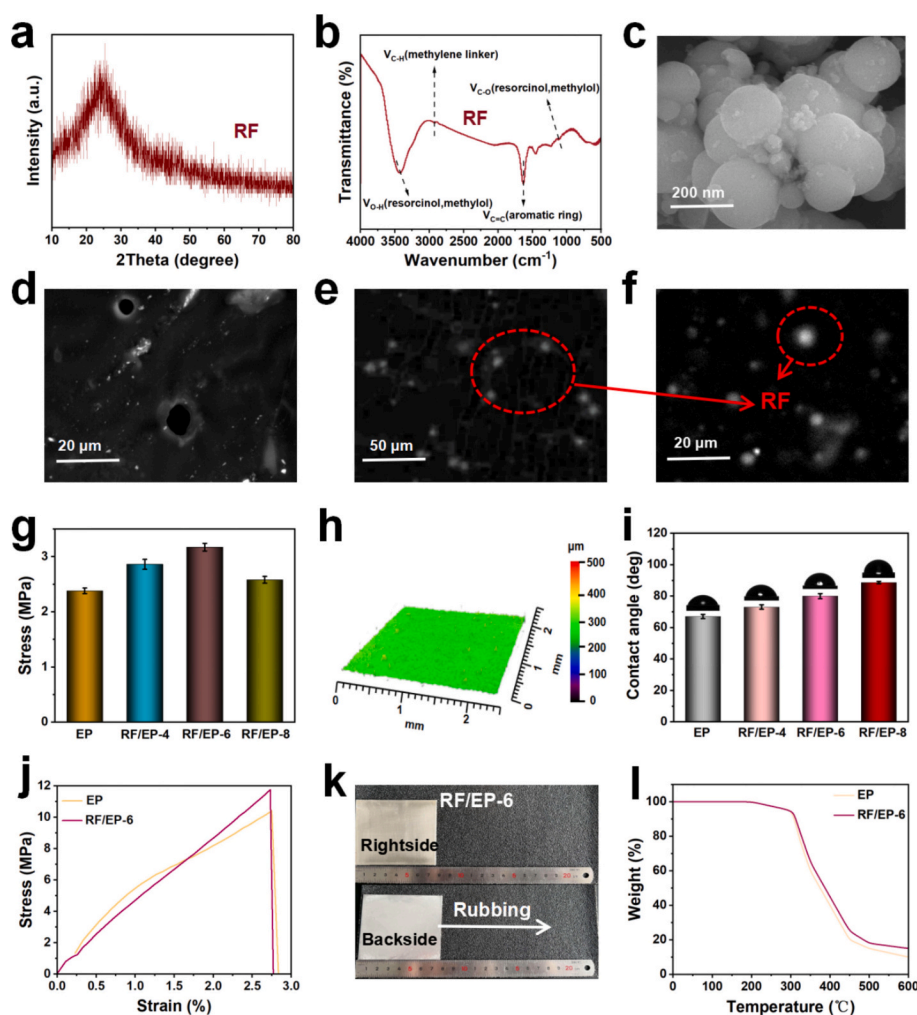
In addition, photocatalytic  $H_2O_2$  production experiments are provided in the Supporting Information.

### 3. Results and discussion

#### 3.1. Structure and properties of RF/EP composite coating

The RF resin was prepared by a simple solvothermal route, and its crystal structure and appearance were characterized by a series of tests. As shown in the X-ray diffraction (XRD) pattern (Fig. 1a), the RF peak at  $2\theta \approx 23^\circ$  presents a broad diffraction signal, corresponding to the graphitic carbon (002) plane caused by  $\pi$ -stacking interactions between electron donor and acceptor units [15]. In Fig. 1b, the Fourier transform infrared (FT-IR) spectrum of RF resin clearly displays the characteristic bands of resorcinol, linker and residue groups, and the characteristic  $\nu_{\text{O-H}}$  oscillatory peaks of methanol or resorcinol can be observed at  $3353 \text{ cm}^{-1}$ , while the characteristic  $\nu_{\text{C-O}}$  oscillatory peaks of both resorcinol and methanol can be found at  $1096 \text{ cm}^{-1}$  [16]. Moreover, the  $\nu_{\text{C-H}}$  and  $\nu_{\text{C=C}}$  oscillation peaks at  $2921$  and  $1610 \text{ cm}^{-1}$  are ascribed to the methylene linker and the aromatic ring, respectively [17]. Fig. 1c presents the scanning electron microscope (SEM) image of RF resin, and it possesses a spherical morphology with a uniform size of  $\sim 500 \text{ nm}$  in average diameter and distribution. It is generally appreciated that EP coating is often used as anti-corrosion coatings in the commercial sector, but most EP coatings on the market today form holes, cracks and other defects during the curing process (Fig. 1d). Thus, the as-prepared RF resin as the filler was mixed with the EP coating to form the RF/EP composite coating. By comparing the RF/EP composite coating (Fig. 1e

and f) with the EP coating, we observed that the surface defects of the RF/EP composite coating were significantly reduced. The RF spherical particles are uniformly dispersed so that the pores are effectively plugged. This creates an effective “labyrinth effect”, which prevents corrosion factors from contacting the metal substrate. In addition, the filling of defects by RF resin leads to a significant reduction of defects in RF/EP composite coatings, which also promotes the densification of RF/EP-6 composite coatings. In general, high adhesion between the Q235 steel plate and the coating plays a positive role in the anti-corrosive coating, and in order to investigate the adhesion properties of the as-prepared coating to the Q235 steel, an adhesion strength test was carried out (Fig. S4). Fig. 1g demonstrates the adhesion ability of each coating, from which it can be seen that the adhesion strength of the RF/EP-6 composite coating is 33.2 % higher than that of the pure EP coating. The increase in the adhesion of the RF/EP-6 composite coating may be mainly due to the incorporation of the RF resin, which is rich in hydroxyl groups, and the hydrogen bonding in these hydroxyl groups creates intermolecular interaction forces between the coating and the surface of the Q235 plate thus improving the adhesion of the composite coating [18]. It is essential to note that the adhesion of fouling organisms like diatoms depends on surface roughness and morphology, which significantly impact coating performance [19]. Therefore, laser confocal scanning microscopy was employed to assess the surface roughness of EP and RF/EP-6 composite coatings (Fig. S5 and Fig. 1h). After adding RF resin, the surface roughness of the composite coating increases, which is



**Fig. 1.** (a) XRD pattern, (b) FT-IR spectrum and (c) SEM image of RF resin. SEM images of the (d) EP coating and (e and f) RF/EP-6 composite coating with different amplification scales. (g) Bonding stress of the coated samples. (h) Surface roughness of the RF/EP-6 composite coating observed by laser confocal scanning microscope. (i) Contact angle data, (j) stress-strain curves, (k) abrading test and (l) TGA curves of the coated samples.

due to the spherical shape of the RF resin, which can fill in the original defects of the EP coating and increases the effective area for interfacial light absorption by forming irregular bumps on the surface of the composite coating. Most test methods for evaluating the wettability of coatings rely on measuring the contact angle between the liquid and the surface of the coating. In general, a contact angle of  $<90^\circ$  indicates that the coating is hydrophilic, i.e., has a high degree of surface wettability, and the contact angle can be measured by droplet shape analysis [20]. Fig. 1i illustrates the results of the contact angle measurements, for several of the coatings measured exhibited hydrophilicity, and the RF/EP-6 composite coating exhibited a favorable contact angle value of  $80^\circ$ , which can be attributed to its high adhesion and low surface roughness—two critical factors influencing the contact angle. To prevent physical damage during service, tensile tests were conducted to evaluate the mechanical properties of the coating. The stress-strain curve (Fig. 1j) demonstrates that EP has a tensile strength of 10.41 MPa and an elongation at a break of 2.86 %. With the incorporation of RF, the tensile

strength of the RF/EP-6 composite coating increased to 11.76 MPa while maintaining an elongation at break close to 2.8 %. This indicates that the reinforcement within the existing crosslinking network was achieved by the introduction of RF, which not only did not affect the original crosslinking sites but also filled the original defects, thereby improving the crosslinking density of the system. To test the wear resistance of RF/EP-6 composite coating, A4 sandpaper No.50 was fixed on the test bench with the sandpaper facing upward, and a 1 kg weight was used to press the coating specimen for grinding (Fig. 1k), with a distance of 5 cm for each grinding, and a total of 20 grinding times. The microstructure of the surface after grinding is provided in Fig. S6, and the average roughness is  $5\text{ }\mu\text{m}$  before grinding and  $2.5\text{ }\mu\text{m}$  after grinding, with little change, which indicates that the composite coating possesses efficient abrasion resistance. The thermal stability of coatings was assessed via thermogravimetric analysis (TGA), and the corresponding curves are displayed in Fig. 1l. Thermal decomposition occurred primarily between  $100^\circ\text{C}$  and  $450^\circ\text{C}$ , with slight mass loss from  $100^\circ\text{C}$  to  $300^\circ\text{C}$  due to

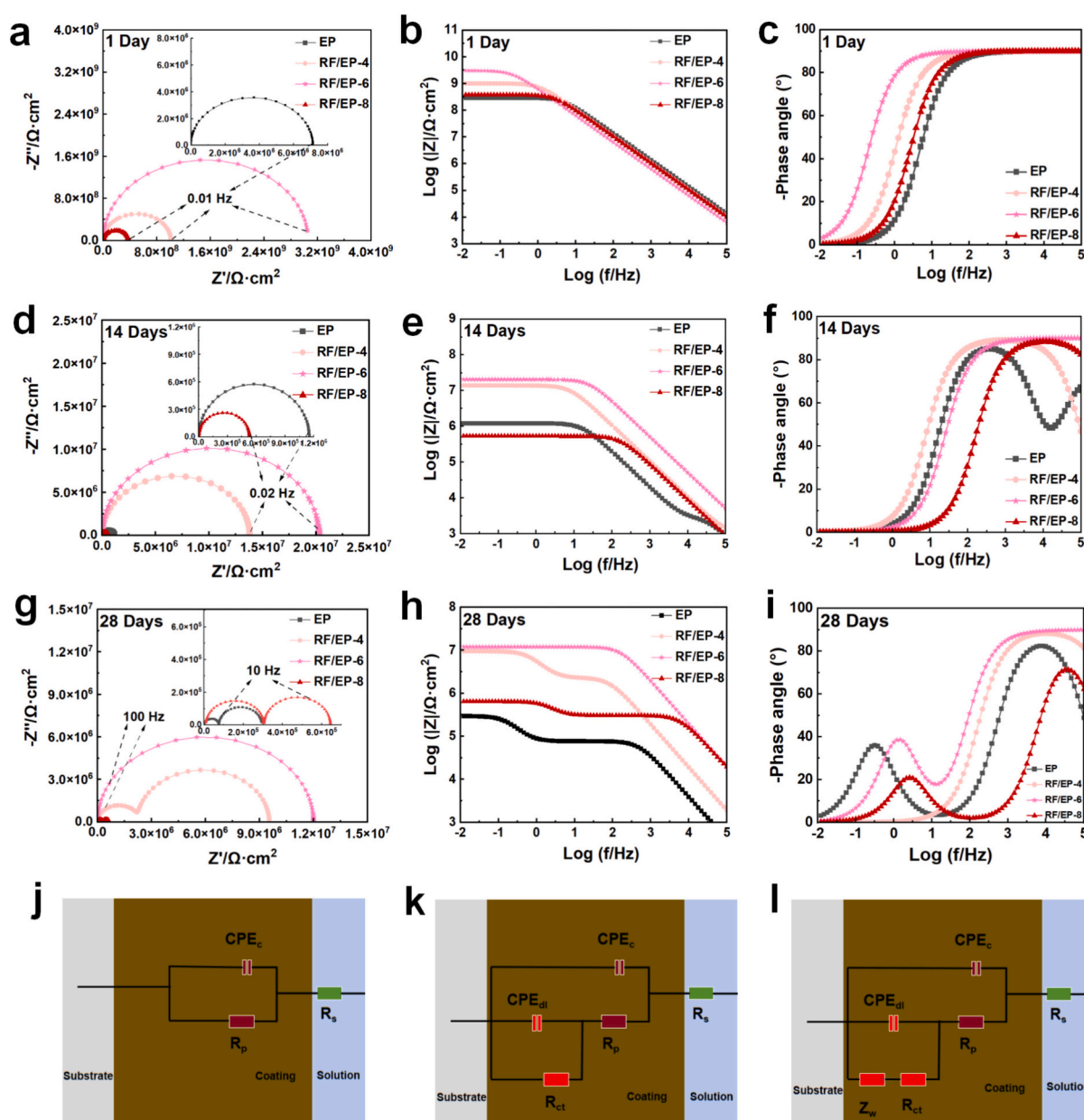


Fig. 2. Nyquist (a, d and g), Bode (b, e and h), and Phase angle (c, f and i) plots for the as-prepared coatings during different immersion times (1, 14 and 28 days). (j–l) Equivalent circuit models used to fit the EIS data.

desorption of small organic molecules [21]. Between 300 °C and 450 °C, significant decomposition and polymer network cracking occurred, primarily involving bisphenol A [22]. The maximum decomposition temperature of the composite coating was about 310 °C, indicating that the RF/EP composite coating remains excellent thermal stability.

### 3.2. Anti-corrosion resistance of RF/EP coatings

Open circuit potential (OCP), often referred to as self-corrosion potential, is commonly used to assess the likelihood of corrosion occurrence, a higher OCP value indicates that the coating is less prone to corrosion and exhibits stronger corrosion resistance [23]. As exhibited in Fig. S7, the EP coating has the most negative OCP value, reflecting its poor corrosion resistance. This is due to the formation of numerous micropores during the curing process, which facilitates the penetration of corrosion ions and water. In contrast, RF/EP composite coatings display higher, more positive OCP values and superior corrosion resistance. The corrosion resistance of the prepared composite coatings was further determined by electrochemical impedance spectra (EIS) measurements, and Fig. 2a–f demonstrate the Nyquist plots, Bode plots, and phase angle plots of the prepared coatings after immersion in 3.5 wt% NaCl solution for different periods. In general, changes in capacitive arc, impedance modulus, and phase angle under low-frequency conditions effectively reflect the coatings' corrosion resistance [24–26]. The RF/EP-6 composite coating was able to achieve impedance values of  $1.56 \times 10^9 \Omega\text{-cm}^2$  even after one day of immersion in the 3.5 wt% NaCl solution, which is comparable to the newly reported properties of the EP-based coatings in 2025 [27]. It is noteworthy that after 28 days of immersion (Fig. 2g–i), the impedance of RF/EP-6 composite coating still exceeds  $6 \times 10^6 \Omega\text{-cm}^2$ , which indicates excellent corrosion-resistant stability with effective barrier performance against corrosive media. In addition, in the discussion of EIS testing it is necessary to refer to the break point frequency ( $f_b$ ) of the coating, i.e. the frequency at which the capacitive behavior changes to a resistive behavior, which can be determined from the phase angle  $-45^\circ$  [28,29]. The  $f_b$  generally demonstrates the degree of coating delamination at the interface between the coating and the metal [30]. From Table S1, it can be seen that the break point frequencies of all coatings increase to different degrees with increasing immersion time in NaCl solution, with the EP coating and the RF/EP-8 composite coating showing a significant increase in frequency, reaching higher frequency values after three weeks of immersion. Surprisingly, the RF/EP-4 composite coating and RF/EP-6 composite coating showed that their break point frequency enhancement levelled off during the immersion process. In particular, the RF/EP-6 composite coating has a frequency of only 1.88 Hz at the end of the three-week immersion, and this behavior indicates that the RF/EP-6 composite coating has very excellent barrier properties. To gain deeper insights into corrosion behavior, EIS data were analyzed using equivalent circuits, as displayed in Fig. 2j–l. The circuit models include  $R_s$ ,  $R_p$ ,  $R_{ct}$ ,  $CPE_c$ ,  $CPE_{dl}$ , and  $Z_w$ , representing the solution resistance, coating resistance, charge transfer resistance, coating capacitance, double-layer capacitance, and Warburg impedance, respectively [31]. At the beginning of immersion, NaCl solution will pass through the micropores created during the curing process of the precast coating, but does not penetrate the contact interface between the coating and the metal substrate, which is consistent with the equivalent circuit provided in Fig. 2j. Then, as the immersion time increases, the contact interface between the coating and the metal substrate is penetrated by corrosion ions, resulting in the formation of a large region of corrosion response, which exhibits a 2-fold constant, consistent with the equivalent circuit presented in Fig. 2k. Finally, the accumulation and diffusion of corrosion products influence the corrosion rate over time. In coatings with poor protection, Warburg impedance may appear during prolonged immersion, as depicted in Fig. 2l.

The corrosion resistance of the different coated samples was evaluated even further by salt spray test, and Fig. 3 reflects the images of the

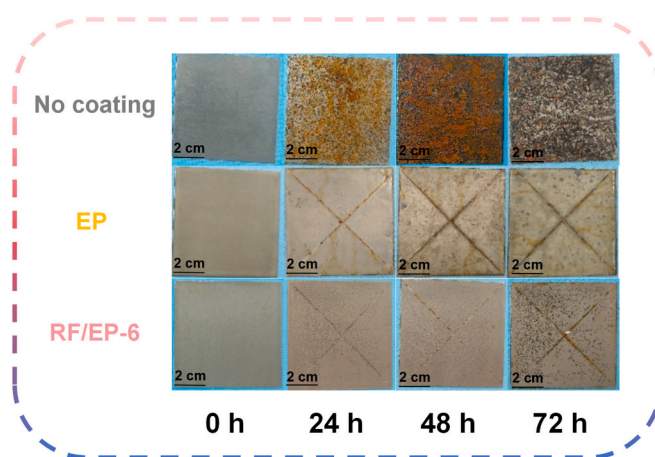
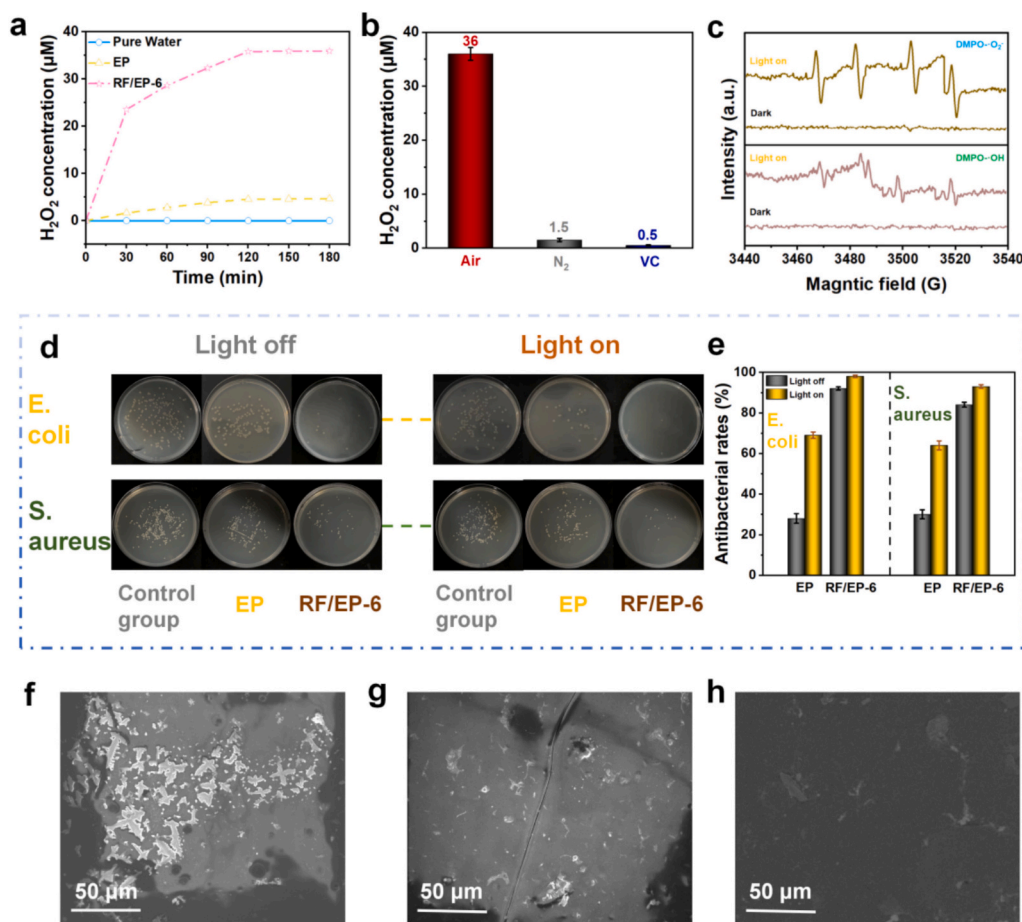


Fig. 3. Optical photographs of the Q235 steel plate with different coatings after the salt spray test for varied time courses.

coated samples after different times of salt spray treatment. After the treatment, the EP-coated Q235 steel plate indicated localized corrosion after 24 h and almost complete corrosion after 48 h, while the uncoated Q235 steel plate was already completely corroded after 24 h. In contrast, the surface of the Q235 steel plate coated with the RF/EP-6 composite coating demonstrated almost no visible corrosion at 24 h. After 48 h, the surface of the Q235 steel plate coated with the RF/EP-6 composite coating began to corrode, indicating that  $\text{Cl}^-$  had penetrated the coating. Significantly, this sample has not shown extensive surface corrosion. These phenomena indicate that the coating with RF incorporation is denser, reducing defects such as holes and cracks, and greatly improving a range of properties of the coating. Interestingly, the Q235 steel plate coated with the RF/EP-6 composite coating showed no serious surface corrosion even after 72 h, while the other samples displaced various degrees of damage. It can be seen that the RF/EP-6 composite coating provides the best surface corrosion protection for the Q235 steel plate, which is in agreement with the results of the electrochemical tests.

### 3.3. Anti-biofouling capacity of RF/EP coating

Ship hulls and offshore equipment are often troubled by various kinds of marine microbial contamination, increasing their maintenance and fuel consumption costs. Consequently, an excellent coating should have both anti-corrosion and anti-biofouling functions to deal with the above problems. In previous literature reports [32], RF resin was able to produce large amounts of  $\text{H}_2\text{O}_2$  during light irradiation, as a green oxidant, which can effectively handle the attachment of Marine micro-organisms [33]. In this regard, the photocatalytic  $\text{H}_2\text{O}_2$  generation test was employed to investigate the photocatalytic performance of as-prepared coatings, as demonstrated in Fig. 4a. It can be seen that almost no  $\text{H}_2\text{O}_2$  was produced by the control group (pure water without adding sample) or EP coating after 3 h of simulated sunlight irradiation, while RF/EP-6 coating can produce  $36 \mu\text{M H}_2\text{O}_2$ , which is ascribed to the unique structure of  $\pi$ -conjugated and  $\pi$ -stacked benzoquinone donor-resorcinol pairs that are capable of catalyzing  $\text{H}_2\text{O}$  oxidation and  $\text{O}_2$  reduction via photo-generated charges, resulting in  $\text{H}_2\text{O}_2$  production [34,35]. To verify that RF resin enables  $\text{H}_2\text{O}_2$  production through two pathways (water oxidation and oxygen reduction), we investigated the photocatalytic  $\text{H}_2\text{O}_2$  production by the RF/EP coating under varying environmental conditions (Fig. 4b). Under a nitrogen environment, minimal  $\text{H}_2\text{O}_2$  was detected. In ambient air, the photocatalytic  $\text{H}_2\text{O}_2$  production in the RF/EP-6 coating primarily proceeds through the oxygen reduction pathway, underscoring the critical role of dissolved oxygen. To further determine that the RF/EP-6 composite coating is mainly an ORR reaction in the photocatalytic production of  $\text{H}_2\text{O}_2$ , electron spin resonance (ESR) radical detection was also carried out (Fig. 4c). The



**Fig. 4.** (a) Photocatalytic  $\text{H}_2\text{O}_2$  generation curves of the control group, EP and RF/EP-6 coatings. (b) Photocatalytic  $\text{H}_2\text{O}_2$  production for controlled experimental conditions of the RF/EP-6 coating under light irradiation (VC: vacuum). (c) ESR spectra of  $\text{DMPO} \cdot \text{O}_2^-$  and  $\text{DMPO} \cdot \text{OH}$  over the RF/EP-6 coating under darkness and light irradiation. (d) Digital photographs of antibacterial tests over the control group, EP and RF/EP-6 coatings. (e) Antibacterial rates of the control group, EP and RF/EP-6 coatings for *E. coli* and *S. aureus*. Images of Chlorella adhered on the (f) control group, (g) EP and (h) RF/EP-6 coatings.

signal corresponding to  $\text{DMPO} \cdot \text{OH}$  of the RF/EP-6 composite coating is almost unobservable, in contrast, the signal of  $\text{DMPO} \cdot \text{O}_2^-$  over the RF/EP-6 composite coating shows a clear signal, which indicates that the RF/EP-6 composite coating mainly produces  $\text{H}_2\text{O}_2$  by the ORR reaction pathway. In the marine environment, bacteria are recognized as one of the major biofouling pathways, which produce secretions over time to create a favorable environment for growth and cause irreparable damage to materials [36,37], so excellent antimicrobial properties are one of the important factors to improve the anti-biofouling effect of coatings. In this study, *E. coli* and *S. aureus* were used to assess the antimicrobial properties by counting the colonies attached to the coatings. The number of colonies attached to the prepared coatings by both the bacteria is getting less and less as shown in Fig. 4d, which is the result of comparison with the control group. Here the control group was also utilized to calculate the bacterial inhibition rate on the prepared coatings and the results are given in Fig. 4e. After the addition of simulated light, the EP coatings showed 69 % and 64 % inhibition against *E. coli* and *S. aureus* respectively. This is due to the antimicrobial properties of the amino groups in the EP coating, which are partly not involved in the curing reaction and can destroy some bacteria [38]. In contrast, the RF/EP-6 composite coating exhibited a higher inhibition efficiency of 98 % and 93 % against *E. coli* and *S. aureus*. Under light conditions, the RF/EP-6 composite coating generates electron-hole pairs. These pairs react with adsorbed water and oxygen to produce reactive species ( $\cdot\text{OH}$  and  $\cdot\text{O}_2^-$ ), which disrupt bacterial membrane permeability. This process degrades bacterial proteases, amino acids, and nucleic acids, ultimately causing cell death [39]. Additionally, the RF/EP-6 composite coating can

generate  $\text{H}_2\text{O}_2$  through the oxygen reduction reaction (ORR). As a potent oxidant,  $\text{H}_2\text{O}_2$  induces ionization of bacterial cellular components (e.g., molecules or atoms), leading to lipid chain breakage in the cell wall and subsequent structural destruction. Algae is also a kind of biological pollution that cannot be ignored in the marine environment, accordingly, Chlorella was selected to examine the resistance of coating to algal adhesion in this study. This can be seen in Fig. 4f–h, less Chlorella attached to the surface of the EP coating compared to the control, because Chlorella tends to attach to hydrophobic surfaces [40], while the EP coating is hydrophilic and can slightly resist the attachment of Chlorella. In our experiments, the addition of RF resin significantly reduced the attachment density of Chlorella on the RF/EP-6 composite coating compared to the EP coating, which is comparable to the newly reported properties of the antifouling coating in 2024 [41]. This is because the RF/EP-6 composite coating can affect the growth of algae by generating reactive species and  $\text{H}_2\text{O}_2$  in the presence of light, water, and  $\text{O}_2$ , thereby destroying the cell wall. Besides, the RF/EP-6 composite coating can make the coating resistant to Chlorella by reducing the adhesion of bacteria and inhibiting the formation of biofilm. Results of anti-algae experiments demonstrated a trend consistent with antimicrobial findings, confirming the outstanding anti-biofouling properties of RF/EP-6 composite coating.

#### 3.4. Optical properties and photogenerated charge of RF resin

A series of photoelectrochemical characterization tests were conducted to further investigate the reason why the unique properties

imparted by the RF resin to the composite coatings resulted in better anti-corrosion and anti-biofouling resistance of the coatings. As reflected in Fig. 5a, the RF resin displays excellent light absorption ability, which helps to enhance the material to produce more photo-generated carriers for anodic protection and more active species for microbial inactivation. In addition, the band gap ( $E_g$ ) value of the RF resin can be calculated to be 2.1 eV from the Kubelka-Munk formula [42,43], and according to VB-XPS result (Fig. 5b), the band structure is presented in Fig. 5c, which indicates that the appropriate energy band structure of RF resin fully satisfies the thermodynamic conditions for the photocatalytic oxygen reduction to produce  $H_2O_2$  ( $O_2/H_2O_2 = 0.68$  V vs NHE). In this regard,  $H_2O_2$  generation in RF resin has a positive effect on the bio-pollution prevention ability of the RF/EP composite coating. Meanwhile, the transient photocurrent response test (Fig. 5d) reveals that the RF resin exhibits strong photocurrent intensity. This demonstrates its capability to efficiently separate photo-generated charges and supply sufficient electrons to the Q235 steel plate surface, enabling cathodic polarization. Such a process facilitates highly efficient anti-corrosion performance in the composite coatings [44,45].

### 3.5. Anti-corrosion and anti-biofouling mechanism

Incorporating RF resin into the EP coating significantly enhances both anti-corrosion and anti-biofouling performance of the resulting RF/EP composite coating. The potential mechanisms behind these enhancements are depicted in Fig. 6. When the surface of the RF/EP composite coating is exposed to sunlight, the RF resin is activated, generating a significant number of photo-induced electron-hole pairs. Some of these photo-generated electrons migrate to the surface of the Q235 steel plate, where they provide cathodic protection, thereby increasing the corrosion resistance of the coating. Moreover, the RF resin within the coating disrupts the pathway for corrosive ions, enhancing the physical barrier effect and bolstering the anti-corrosion performance of the coating. In addition, the electron-hole pairs excited by light facilitate the reaction between  $H_2O$  and  $O_2$ , resulting in the

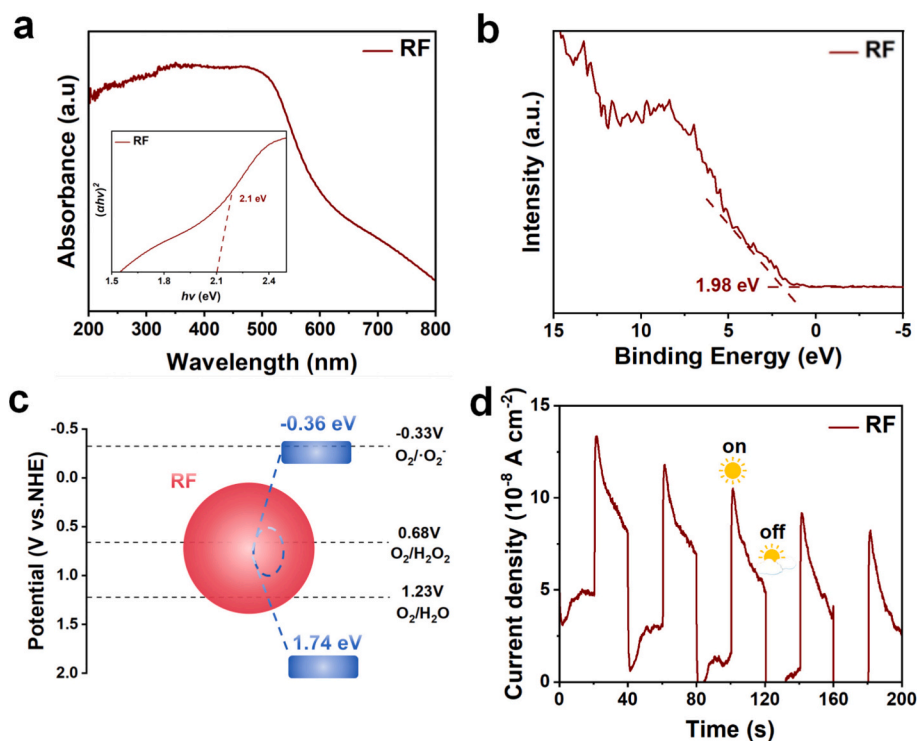
formation of active substances such as  $\bullet OH$  and  $\bullet O_2^-$ . Through the oxygen reduction reaction (ORR), these substances generate  $H_2O_2$ , which exhibits a significant inactivation effect on free microorganisms in seawater and prevents their attachment. The anti-biofouling properties imparted to the coating indirectly improve the anti-corrosion resistance of the Q235 carbon steel substrate.

## 4. Conclusion

In summary, the successful introduction of RF resin in EP coatings to form RF/EP composite coating positively impacts the anti-corrosion and anti-biofouling resistance of Q235 steel plates. EIS tests demonstrated that the optimum RF/EP-6 composite coating presented a two-order of magnitude increase in impedance radius compared to the pure EP coating, proving the excellent outstanding anti-corrosion resistance. The RF/EP composite coating also indicated outstanding resistance to biological contamination through antimicrobial and anti-algae testing. The superior performance stems primarily from (i) RF resin's ability to fill micropores in the EP coating, thereby enhancing the barrier effect and improving corrosion resistance; (ii) RF resin can provide enough photo-excited electrons to the surface of Q235 steel plate to achieve cathodic polarization under sunlight; and (iii) the RF resin can produce active species ( $\bullet OH$  and  $\bullet O_2^-$ ) and  $H_2O_2$ , which effectively prevent marine microorganisms from the attachment and inactivation of coating. Most of the marine coatings currently reported in the market have only anti-corrosion properties, which are monofunctional, cannot be adapted to the marine environment for a long period of time and are relatively complicated to prepare. This work is the first to use RF resin as a novel filler in combination with EP coatings in the field of corrosion prevention and resistance to microbial contamination, expanding its development and application in durable marine equipment coatings.

### CRediT authorship contribution statement

Yuanhao Tang: Writing – original draft, Data curation,



**Fig. 5.** (a) UV-vis diffuse absorption spectrum (inset shows band gap energy), (b) VB-XPS spectrum, (c) corresponding energy band structure and (d) transient photocurrent response of RF resin.

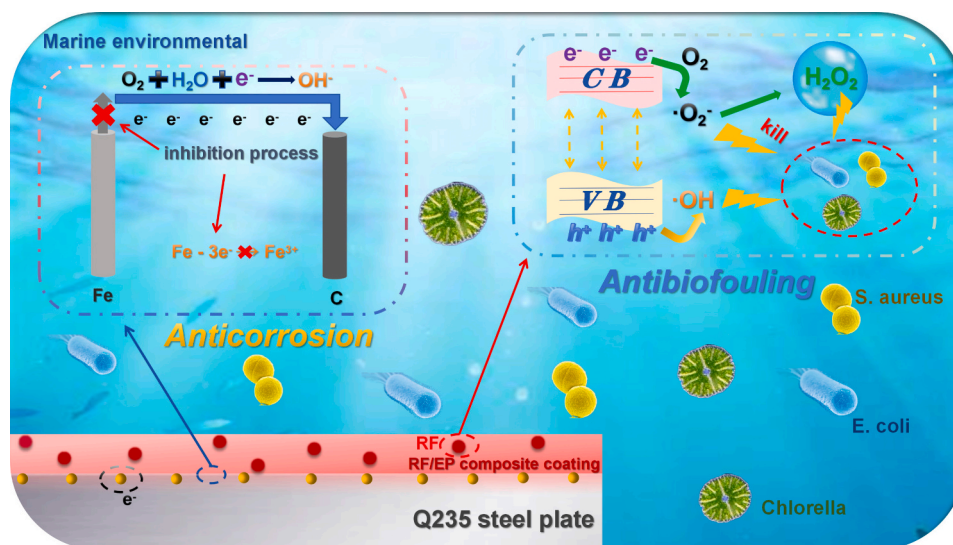


Fig. 6. Possible anti-corrosion and anti-biofouling mechanism of the RF/EP composite coating.

Conceptualization. **Xiaojie Wu**: Data curation. **Suchang Zou**: Data curation. **Haoyuan Qin**: Data curation. **Kaiqu Sun**: Data curation. **Lei Sun**: Data curation. **Jianhua Hou**: Data curation. **Weilong Shi**: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Chunsheng Li**: Resources, Project administration, Funding acquisition. **Feng Guo**: Resources, Project administration, Funding acquisition.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2025.109177>.

#### Data availability

Data will be made available on request.

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